

Topics in this lecture

- Sample space
- Events *เหตุการณ์*
- Probability *ความน่าจะเป็น*

Why Learn Prob & Stat?

- Give you the power to model things that are random in nature
- Can be applied in many fields of study
 - Science
 - Engineering
 - Business

Motivating Example 1

- Netflix grand prize
 - Movie recommendation system
 - Predicts how a viewer will rate a movie based on his/her ratings
- \$1,000,000 prize!!
(Awarded in 2009)



Motivating Example 2

- Performance modeling
 - How many cashiers do I need in order to keep the average waiting time under 5 minutes?



Motivating Example 3

- Classification
 - How does a spam filter know which e-mail is a spam?

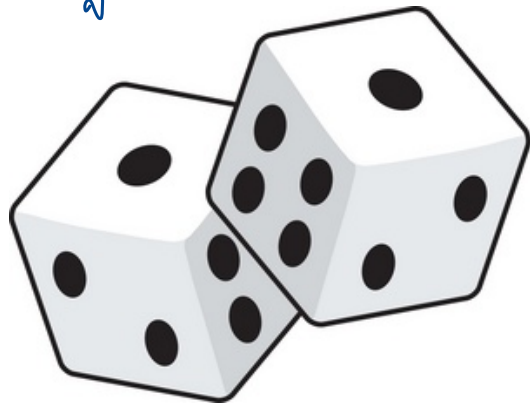


การทดลองสุ่ม

Random Experiments

- Outcome varies in an *unpredictable fashion* when the experiment is repeated under the same conditions
- Must specify clearly what to be measured or observed

โยน 1 ลูก
ผลลัพธ์ที่ได้อาจเป็นอะไรก็ได้



Experiment 1

Select a ball from an urn containing balls numbered 1 to 50. Note the **number of the ball**

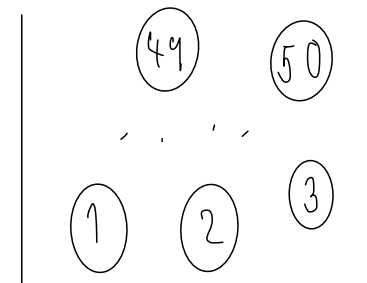
๑. ค่าจะเป็นที่อะไร #บอลเลขที่

$$P(t_1) = \frac{25}{50} = \frac{1}{2} \#$$

Possible outcomes:

1, 2, 3, ...

$$\Omega = \{1, 2, 3, \dots, 50\}$$



Experiment 2

เลือกบอล 1 ลูก

Select a ball from an urn containing balls numbered 1 to 4. Suppose balls 1 and 2 are black and balls 3 and 4 are white. Note the **number and color** of the ball selected.

$$E_2 = \{ (4, W) \}$$

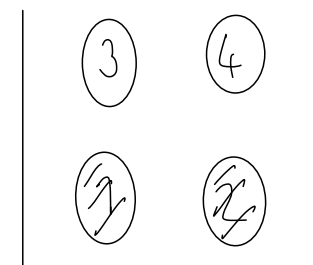
$$P(E_2) = \frac{1}{4} \#$$

๑. ถ้าจะเลือกที่จะได้
บอลขาวที่ ๔

Possible outcomes:

$(3, \text{white}), (2, \text{black})$

$$\Omega = \left\{ \begin{array}{l} (1, B), (2, B) \\ (3, W), (4, W) \end{array} \right\}$$



Experiment 3

โยนเหรียญ 3 ครั้ง

จดลำดับของ H, T ที่ออกหลังจาก 3 ครั้ง

Toss a coin **3 times** and note a **sequence of heads and tails**

$$E_3 = \{ HHH, TTT \}$$

$$P(E_3) = \frac{2}{8} = \frac{1}{4} \neq$$

Possible outcomes:

HHH, THT, ..., TTT

$$\Omega = \left\{ \begin{array}{l} HHH, HHT, HTH, THH \\ TTT, TTH, THT, HTT \end{array} \right\}$$

Experiment 4

โยน 3 เหรียญ

ออกหัวกี่ครั้ง

Toss a coin 3 times and note the number of heads

$$E_4 = \{ \}, \emptyset \quad (\text{โยนไปโยนมา = get nothing})$$

Possible outcomes:

0, 1, 2, 3

$$\Omega = \{0, 1, 2, 3\}$$

$E_{4.1}$ โยน 3 เหรียญ ออกหัว 3 ครั้ง

$$E_{4.1} = \{0, 3\}$$

$$P(E_{4.1}) = \frac{2}{4} = \frac{1}{2}$$

Experiment 5

A packet is transmitted repeatedly over a noisy channel until it is received without error at the receiver. **Count the number of transmissions required.**

ส่งซ้ำๆ จนกว่าจะส่งสำเร็จ



Possible outcomes:

1, 2, 3, ...

$$\Omega = \{1, 2, 3, \dots, 100, \dots\}$$

$$\mathcal{E}_s = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$$

Experiment 6

สุ่มเลือกตัวเลข 1 ตัว ที่อยู่ระหว่าง 0 กับ 1

Pick a number at random between 0 and 1.

$$E_{6.1} = \text{เลขที่สุ่มได้ 1 ตัว อยู่ระหว่าง 0-3}$$

$$= \{x : 0 < x < 0.3\}$$

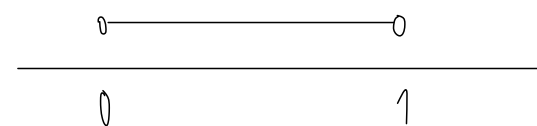
$$P(E_{6.1}) =$$

Possible outcomes:

0.5, 0.1, 0.919, 0.0000000001, ... # เขียนออกมาได้ ๓

$$\Omega_6 = \{0 < x < 1\}$$

เลขทุกตัวไม่ 0 ถึง 1 เขียนเป็นช่วง



$$E_6 = \Omega_6$$

Experiment 7

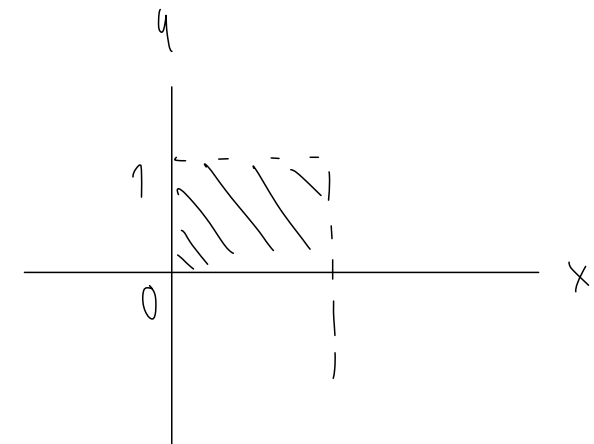
Pick two numbers at random between 0 and 1.

$$\Omega = \{ (x, y) : 0 < x < 1, 0 < y < 1 \}$$

Possible outcomes:

$$(0.5, 0.012), (0.998, 0.123), \dots$$

$$E_7 = \left\{ (x, y) : |x - y| < \frac{1}{10}, \right. \\ \left. 0 < x < 1, 0 < y < 1 \right\}$$



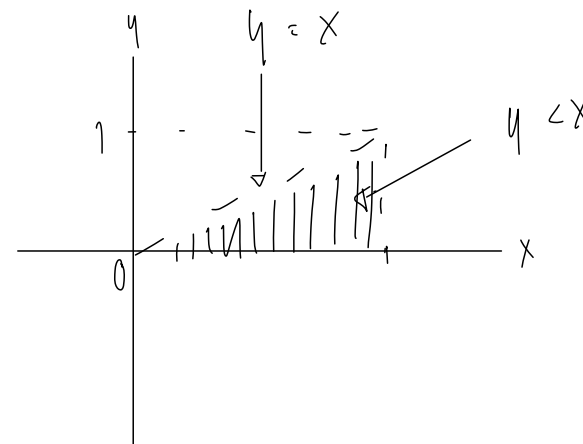
Experiment 8

Pick a number X at random between 0 and 1, and pick a number Y at random between 0 and X .

$$\Omega = \{ (X, Y) : 0 < X < 1, \quad 0 < Y < X \}$$

Possible outcomes:

$$(0.8, 0.2)$$



Sample Space

เปลี่ยน. ทดลอง ทั้งหมดที่ไปไม่ได้
เก็บใน set

- Definition: The sample space S of a random experiment is the *set of all possible outcomes*
- Exercise:
 - Define the sample spaces of Exp1 - Exp8

Types of Sample Spaces

ผลลัพธ์หนึ่ง — นับได้ (แบ่งออกได้)

- **Discrete** — Countable

- Outcome can be put into one-to-one correspondence with positive integers.

- May be finite or countably infinite

นับได้แบบจำกัด, นับได้แบบไม่จำกัด

ต่อเนื่อง

นับไม่ได้ (แบ่งออกไม่ได้)

- **Continuous** — Uncountable

↳ Ex. ช่วงเวลา ที่วิ่งอยู่ระหว่าง 0 ถึง 1

Events

$$\Omega = \{ \dots, \dots \}$$

6 9 0 7 1 1 7 8 1 2 4

← ผลลัพธ์ที่อาจเกิดขึ้น

$$E = \{ \dots, \dots \}$$

↓

- **Subset** of the sample space

- Special events

เหตุการณ์ที่เกิดขึ้นทั้งหมด = sample space Ω

- **Certain event** — the **sample space** itself, consists of all possible outcomes

เหตุการณ์ที่ไม่เกิดขึ้น

- **Impossible event** — the **null event**, contains **no outcome**

$$\emptyset, \{ \}$$

Example Events of Exp1 - Exp8

- E1 = “An even-numbered ball is selected”
ได้บอลเลขคู่
- E2 = “The ball is white and even-numbered”
สีขาวเลขคู่
- E3 = “The three tosses give the same outcome”
โยน 3 ครั้ง ออกมาเหมือนกันหมด
- E4 = “The number of heads equal the number of tails”
จำนวนหัว = etail
- E5 = “Fewer than 10 transmissions are required”
ส่งน้อยกว่า 10 ครั้ง
- E6 = “The selected number is non-negative”
เลขที่เลือก ไม่ติดลบ
- E7 = “The two number differs by less than 1/10”
เลขสองตัวที่ 1 กับ ตัวที่ 2 กัน ไม่เกิน 0.1
- E8 = “The two number differs by less than 1/10”

Experiments Summary

Experiment	Sample Space	Event
1	$S1 =$	$E1 =$
2	$S2 =$	$E2 =$
3	$S3 =$	$E3 =$
4	$S4 =$	$E4 =$
5	$S5 =$	$E5 =$
6	$S6 =$	$E6 =$
7	$S7 =$	$E7 =$
8	$S8 =$	$E8 =$

Creating Events

- *Elementary events* — events from a discrete sample space which consists of a *single outcome*
- Events can be combined using set operations
- Set operations
 - Union
 - Intersection
 - Complement

Set Arithmetic

- Commutative:
 - $A \cup B = B \cup A$
 - $A \cap B = B \cap A$
- Associative:
 - $(A \cup B) \cup C = A \cup (B \cup C)$
 - $(A \cap B) \cap C = A \cap (B \cap C)$

Set Arithmetic (2)

- Distributive:
 - $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
 - $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
- De Morgan's Rules:
 - $(A \cup B)' = A' \cap B'$
 - $(A \cap B)' = A' \cup B'$

Probability

ความน่าจะเป็น
= ค่าซึ่งที่ กำหนดให้กับผลลัพธ์ของ event

- Probabilities are numbers assigned to events to indicate how “likely” it is that the events will occur
- A probability law is a rule that assigns probabilities to the events of the experiment
- A probability law is a function that *assigns a number to a set (event)*

Simple Probability Assignment

- If the sample space is discrete with n elements and each outcome is *equally likely*, then we can assign each elementary event a probability of $1/n$

$\bar{v}$

หาก n ใด ๆ ใน sample space ต้องมีโอกาสพอ ๆ กัน

$$\frac{1}{n}$$

Example 1.1

- An urn contains 10 identical balls numbered 0, 1, ..., 9. Select a ball and note the number on the ball. Find the probability of the following events:
 - A = “number of ball selected is odd”
 - B = “number of ball selected is multiple of 3”
 - C = “number of ball selected is less than 5”
 - D = “number of ball selected is odd or less than 5”

Example 1.2

- Toss a coin 3 times. Find the probability of getting 2 heads out of 3 tosses.

$$\Omega = \left\{ \begin{array}{l} HHH, HHT, HTH, THH \\ TTT, TTH, THT, HTT \end{array} \right\}$$

$$E_A = \{ HHT, HTH, THH \}$$

$$P(A) = \frac{3}{8} \#$$

நிழலா பரவ இயல்

७७९९ १०११०११

Assigning Probability: Continuous

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- Involve assigning numbers to intervals on the real line and region in the plane
- In case of equally likely outcomes, assign probability based on length of interval or size of region in the plane

Frage Wie wird die Nachfrage von S

event

sample space

size

Example 1.3

$$\Omega = \{0 < X < 1\}$$

- Pick a random number between 0 and 1. Find the probability of the following events.
- A = “the selected number is less than 0.5” $E_A = \{0 < X < 0.5\}$

$$P(A) = \frac{0.5}{1}$$
- B = “the selected number is between 0.3 and 0.6” $E_B = \{0.3 < X < 0.6\}$

$$P(B) = \frac{0.3}{1}$$
- C = “the selected number is negative” $\emptyset = P(C) = 0$

Example 1.4

$$\Omega = \{ (x, y) : 0 < x, y < 1 \}$$

- Pick 2 numbers x and y at random between 0 and 1. Find the probability of the following events

- $A = "x > 0.5"$ $E_A = \{ 0.5 < x < 1 \}$ $P(A) = \frac{0.5}{1}$

- $B = "y > 0.5"$ $E_B = \{ 0.5 < y < 1 \}$ $P(B) = \frac{0.5}{1}$

- $C = "x > y"$ $E_C = \{ y < x < 1 \}$ $P(C) = 1$

ខ្លឹមសារ ៤ ចំណុច

Axioms of Probability

- Axiom 1:

៖ ប្រូបាប៊ីលីតេ គឺជា លេខ វិជ្ជមាន

$$0 \leq P(A)$$

- Axiom 2:

៖ ប្រូបាប៊ីលីតេ នៃ លំហ គឺស្មើ ១

$$P(\Omega) = 1$$

Axioms of Probability (2)

- Axiom 3:

If $A \cap B = \emptyset$, then $P(A \cup B) = P(A) + P(B)$

Corollaries

- Corollary 1:

$$P(A') = 1 - P(A)$$

- Corollary 2:

$$P(A) \leq 1$$

- Corollary 3:

$$P(\emptyset) = 0$$

Corollaries (2)

- Corollary 4:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$